

Algebra 1 Unit 11 Assessment Guide

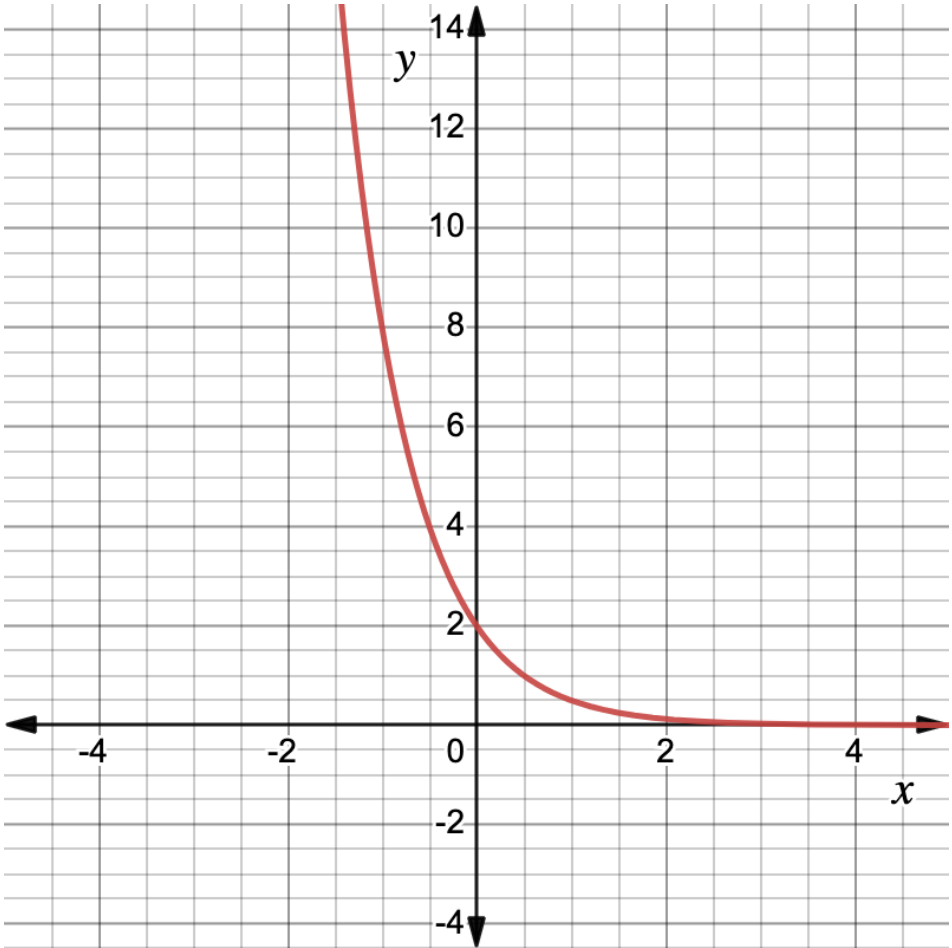
General Information

Unit 11 begins a student's journey in the learning and understanding of exponential functions. Exponential functions were introduced in Unit 1 when students explored parent functions. While students learn about exponential functions, they also find that exponential functions have a multitude of applications. Students continue to explore functions through their key features. The common theme of key features in the benchmarks can help students become better acquainted with exponential functions because they have utilized key features with linear, quadratic, and absolute value functions.

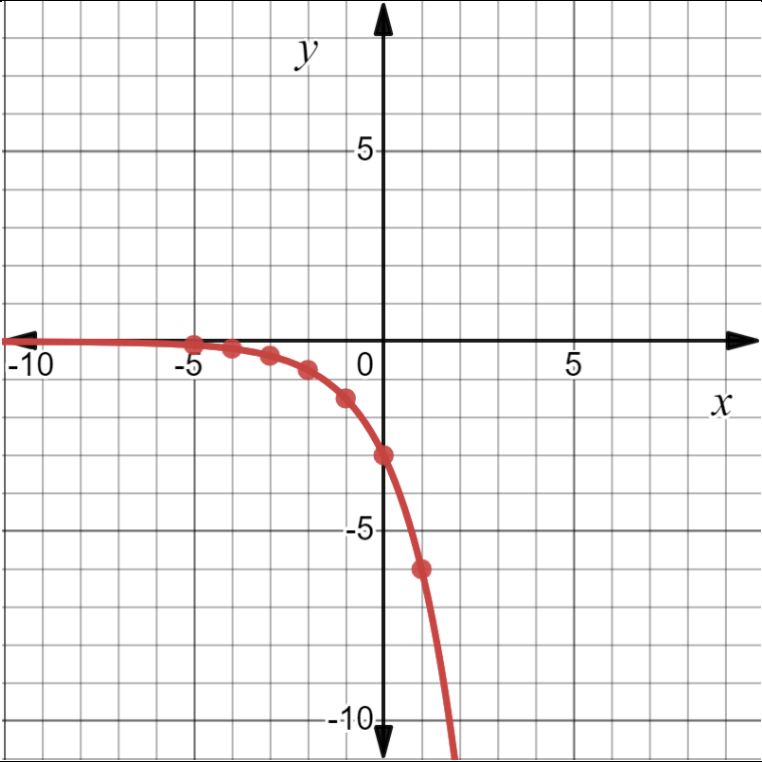
Students may struggle with the constant percent rate of change. Teachers may wish to support students with this concept by using resources from Grade 7, Unit 7, Lesson 9. This lesson is on finding percent increase and percent decrease. Another possible way to help students grasp how a constant percent rate of change is not the same as a linear change is to have them "seed" a calculator (in Desmos this is using the answer function) with the number 100 and to multiply by a b -value such as 1.1 or 0.9. Write the exponential function that accompanies the "seed" and the b -value and then have the student complete a table by taking the result of each multiplication and using the result as the next "seed". The x -value is the number of iterations. The tables shown are examples.

Seed = 100, $b = 1.1$, $y = 100(1.1)^x$		Seed = 100, $b = 0.9$, $y = 100(0.9)^x$	
x	y	x	y
Seed (0 iterations)	100	Seed (0 iterations)	100
1	110	1	90
2	121	2	81
3	133.1	3	72.9

	Teachers may wish to monitor whether students are struggling with certain key features. Since key features such as domain, range, intercepts, intervals for increasing, decreasing, positive, and negative, and end behavior are not new, students who are struggling on these may need further support. Consider monitoring the types of questions that students struggle with when writing the equation of an exponential function. For example, do students struggle more with a graph than a table of values and a written description. This may help further distinguish how best to support students who struggle with MA.912.AR.5.4. Questions 5 and 6 give the percent rate of change using different verbiage. If a student gets one but misses the other, consider if this is because of the change in the wording.
Calculator Information	The questions on the assessment are calculator.
Total Recommended Points	This assessment is a total of 100 points.

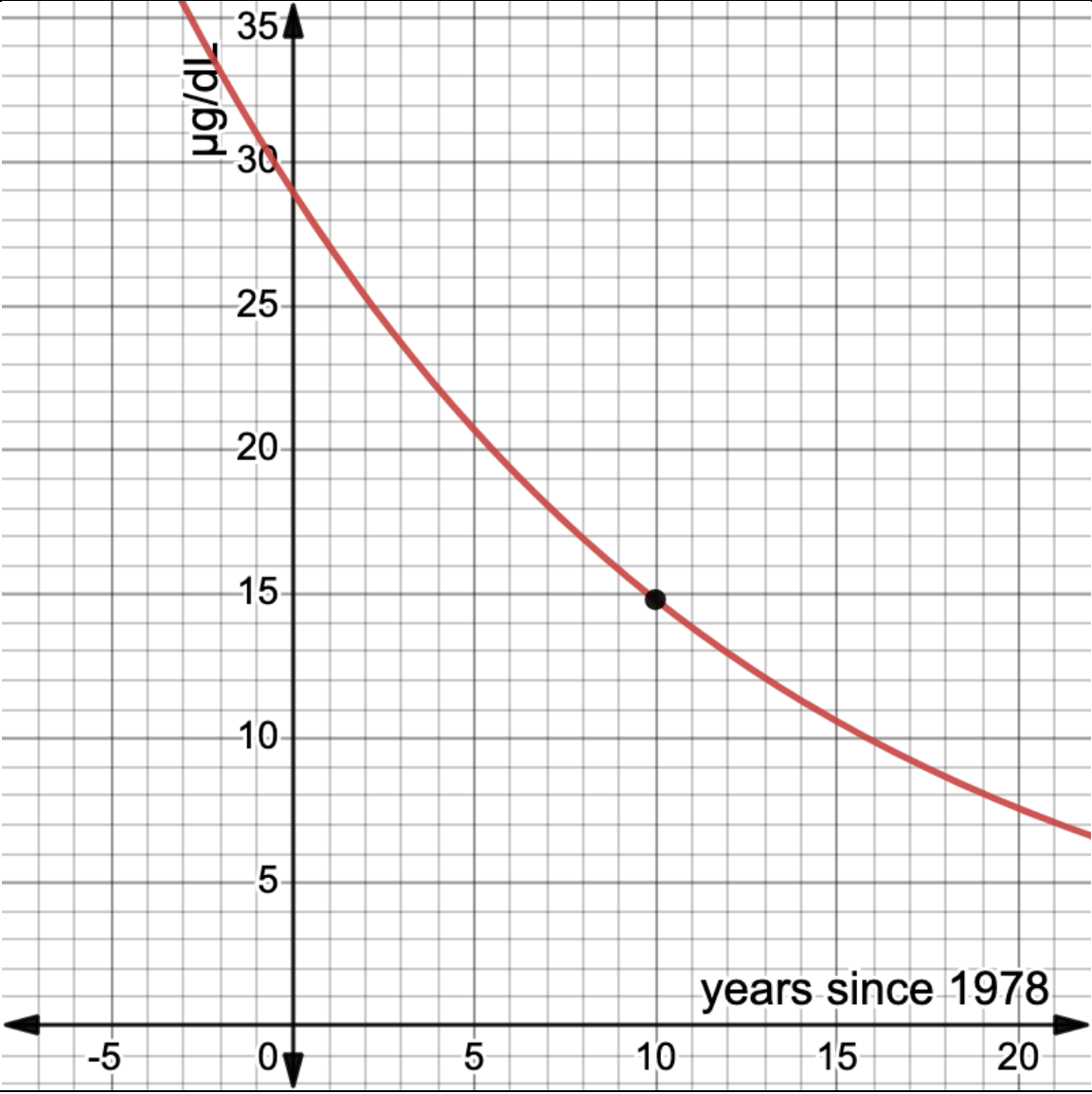
Question	1a	Benchmark(s)	MA.912.AR.5.6
		Recommended Scoring (4 Total Points)	Teachers are encouraged to note how students approached graphing the equation. If students use the equation to generate a table of values, then they may incorrectly written down a value. Students may have determined key features so their preciseness may not be the same as a student who uses a table of values. Teachers should look for a graph that is precise based on the student's method. Therefore, the graph may not contain all points so the curve may not be as smooth as when a student uses several points to graph.

Question	1b				Benchmark(s)	MA.912.AR.5.6																
<table><tr><th>Name</th><th>Key Feature</th><th>Name</th><th>Key Feature</th></tr><tr><td>x-intercept(s)</td><td>none</td><td>increasing</td><td>none</td></tr><tr><td>y-intercept</td><td>(0, 2)</td><td>decreasing</td><td>R</td></tr></table>					Name	Key Feature	Name	Key Feature	x-intercept(s)	none	increasing	none	y-intercept	(0, 2)	decreasing	R	Recommended Scoring (16 Total Points)		Consider distributing the points so that each key feature is 4 points.			
Name	Key Feature	Name	Key Feature																			
x-intercept(s)	none	increasing	none																			
y-intercept	(0, 2)	decreasing	R																			
Question	2				Benchmark(s)	MA.912.AR.5.3																
<table><tr><td></td><td>$f(x) = -\frac{1}{2}(5)^x$</td><td>$g(x) = 4\left(\frac{1}{7}\right)^x$</td><td>$f(x) = (2)^x$</td><td>$f(x) = (1 + 0.3)^x$</td></tr><tr><td>Exponential Decay</td><td>☐</td><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>Exponential Growth</td><td>☒</td><td>☐</td><td>☒</td><td>☒</td></tr></table>						$f(x) = -\frac{1}{2}(5)^x$	$g(x) = 4\left(\frac{1}{7}\right)^x$	$f(x) = (2)^x$	$f(x) = (1 + 0.3)^x$	Exponential Decay	☐	☒	☐	☐	Exponential Growth	☒	☐	☒	☒	Recommended Scoring (16 Total Points)		Consider distributing the points so that each choice is 4 points.
	$f(x) = -\frac{1}{2}(5)^x$	$g(x) = 4\left(\frac{1}{7}\right)^x$	$f(x) = (2)^x$	$f(x) = (1 + 0.3)^x$																		
Exponential Decay	☐	☒	☐	☐																		
Exponential Growth	☒	☐	☒	☒																		

Question	3a	Benchmark(s)	MA.912.AR.5.6
		Recommended Scoring (4 Total Points)	Students should graph the function using the table of values provided.
Question	3b	Benchmark(s)	MA.912.AR.5.4
	$y = -3(1 + 1)^x$	Recommended Scoring (4 Total Points)	Consider giving students partial credit if they correctly wrote the function in the form $y = ab^x$.

Question	4a	Benchmark(s)	MA.912.AR.5.3																						
<table border="1"> <thead> <tr> <th colspan="4">Key Features</th></tr> </thead> <tbody> <tr> <td>Domain</td><td>$\mathbb{R}$</td><td>Range</td><td>$y > 0$</td></tr> <tr> <td>y-intercept</td><td>$(0, 8)$</td><td>Constant Percent Rate of Change</td><td>$\frac{3}{4}$</td></tr> <tr> <td rowspan="2">Increasing or Decreasing</td><td rowspan="2">decreasing</td><td>Positive</td><td>$\mathbb{R}$</td></tr> <tr> <td>Negative</td><td>none</td></tr> <tr> <td>Left End Behavior As $x \rightarrow -\infty$,</td><td>$y \rightarrow \infty$</td><td>Right End Behavior As $x \rightarrow \infty$,</td><td>$y \rightarrow 0$</td></tr> </tbody> </table>		Key Features				Domain	$\mathbb{R}$	Range	$y > 0$	y-intercept	$(0, 8)$	Constant Percent Rate of Change	$\frac{3}{4}$	Increasing or Decreasing	decreasing	Positive	$\mathbb{R}$	Negative	none	Left End Behavior As $x \rightarrow -\infty$,	$y \rightarrow \infty$	Right End Behavior As $x \rightarrow \infty$,	$y \rightarrow 0$	Recommended Scoring (36 Total Points)	Consider distributing the points so that each choice is 4 points.
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Question	4b	Benchmark(s)	MA.912.AR.5.4																						
$y = 8\left(\frac{3}{4}\right)^x$		Recommended Scoring (4 Total Points)	Consider giving students partial credit if they correctly wrote the function in the form $y = a(1 - r)^x$.																						
Question	5	Benchmark(s)	MA.912.AR.5.4																						
$P(t) = 26.5(0.9)^t$		Recommended Scoring (4 Total Points)	Students can choose to write the function in any form. Consider a minor deduction if students use the variables x and y instead of the provided variables although preciseness knowing how to write a function given a written description is the focus of the benchmark.																						

Question	6	Benchmark(s)	MA.912.AR.5.4
	$S(t) = 0.12(1.417)^x$	Recommended Scoring (4 Total Points)	Students can choose to write the function in any format. Consider a minor deduction if students use the variables x and y instead of the provided variables although preciseness knowing how to write a function given a written description is the focus of the benchmark.
Question	7	Benchmark(s)	MA.912.AR.5.4
	$y = 5.8(1.1)^x$	Recommended Scoring (4 Total Points)	Consider giving students 2 points if they estimate the y -intercept using the graph instead of finding it using the percent rate of change.

Question	8	Benchmark(s)	MA.912.AR.5.6
		Recommended Scoring (4 Total Points)	Consider giving students 2 points if they “guess” at the y-intercept instead of finding it using the written information but correctly use the given key features in sketching the graph.